Neural Probes
Course Project, Biomedical Microsystems

Motivation and problem statement:

Psychology, the scientific study of human mind and its functions, with the development of technology and resources has developed an alternate and more direct approach towards the understanding of human brain and nervous system called neurosciences. The quest of developing solutions to neural disorder and our fascination towards the immense computational power of a brain has increased our curiosity towards the functioning of a brain and the nervous system. The primary step involved in neural study is the analysis of spontaneous nerve impulses produced as a response to a stimulus. This presents a need for transducers that can sense and gauge neural impulses with the highest possible accuracy and resolution. Neural probes are devices that do this job.
Neural probes are actively used in investigation of brain diseases, neuroprosthetics, and brain mapping. Focus lies on developing probes that are easy to implant, transmit reliable reading, provide a larger and resolved set of impulse readings over a given area and have better biocompatibility.

Our focus in this project will be on designing a 3D electrode array, which can be implanted on the brain surface to sense neural activity in the region. Each of these electrodes should be thin to ensure minimum damage to neural tissues but capable of holding multiple recording sites. It should also be tough enough to withstand the tension and compression it may experience during implantation and on site working.

Brainstorming Report:

Neural probes are presently being used for capturing neural impulses in the brain and spinal, using electrode (arrays) and also for stimulating the neural networks, either electrically or chemically. Thus, state of the art neural probes provide both electrical and chemical connectivity by employing microelectrodes and micro wires, and micro channels respectively.

In our report, we present three alternate designs for neural probes differing in their structure, fabrication and hence, overall applications.

1. 3D neural probes based on a silicon island structure and a simple folding procedure.
2. Planar 2D-Array Neural Probe for Deep Brain Stimulation and Recording (DBSR).
3. Polymer 3D electrode arrays developed by magnetically bending polymer shanks out of the plane.

<u>Design 1</u>: 3D neural probes based on a silicon island structure and a simple folding procedure.

Salient features of this design:

- Provides a simple and reliable method to fabricate and assemble high density 3D arrays of electrodes.
- Allows integrating micro channels to enable chemical excitation of neural networks.
- Each array consists of multiple silicon islands, folded back to back or face to face.

- Each island has multiple probes (shanks) extending outwards (cantilevers) that carry two-three electrodes and microchannel outlets.
- The fabrication process is also post-CMOS compatible, allowing the monolithic integration of CMOS circuits with neural probes using an economic post-CMOS process.

Abstract of design for the 3D probe presented henceforth:

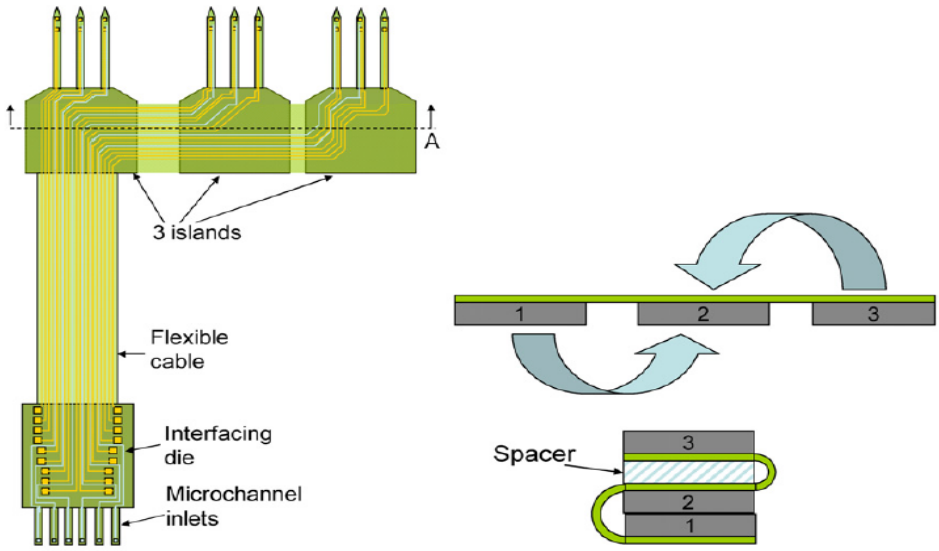


Figure 1: Schematic of the proposed device and assembly method of the 3D probes

The 3D array proposed consists of 3 silicon islands. All the islands are fabricated as a 2D interconnected chain and are linked using the flexible skin process [1]. First, island 1 is folded to island 2 back to back. Next, island 3 is folded to island 2 face to face. Note that a spacer is inserted between islands 2 and 3 to adjust the distance between probes in row 2 and row 3. During this assembly process, epoxy is used to glue all pieces together.

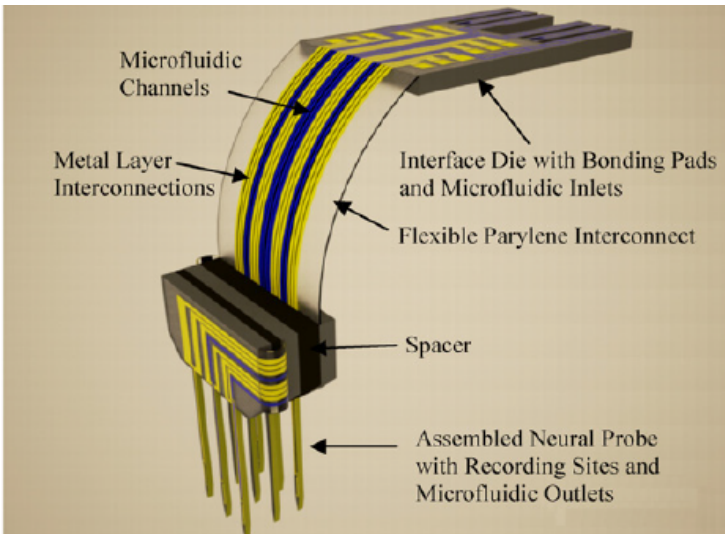


Figure 2: 3D illustration of the assembled neural probes

Each of the islands supports 3 shanks (cantilevers) that will enter the brain cortex. Each shank houses two electrodes and a couple of microfluid outlets. Thus, the total network appears as a 3x3 projection network with a total of 18 electrodes and 18 microfluid outlets.

Fabrication process:

- Fabrication of the neural probe is carried out on a double-side polished silicon wafer. The upper surface is employed for producing the parylene embedded interconnects while the lower surface is developed into silicon islands and probe shanks.
  The fabrication process we referred to made use of a 512 μm thick 5 inch diameter double-side polished silicon wafer.

- First step is the development of an oxide layer (about 300 nm) on the upper wafer surface. We can employ dry oxidation technique wherein Si wafer is exposed to oxygen at 1 atm and 1000^0 C.
  This oxide layer serves as an insulator for the metal electrode will develop next.

- A chrome and gold film needs to be deposited and patterned to generate the metal electrodes and conduction paths.
  We basically employ etch back process. 20 nm chrome and 200 nm gold film is deposited using evaporation based additive transfer. Next, a photoresist is spin-coated and selectively exposed using a suitable mask. The photoresist is then developed to achieve the desired patterns. Next, the gold and chrome layers are etched using gold etch bath and chrome etch bath respectively. Finally, the residual photoresist is dissolved to get the electrode and conduction paths as desired.

- Now, the exposed silicon dioxide layer is etched by Drytek reactive ion etching (DRIE) using C_6F_6.
  A 5 μm thick parylene C layer is deposited and patterned to open to open the access areas for the microchannels (including inlet port and outlet ports). The layer is patterned using oxygen plasma etching with the help of an aluminium mask.

- XeF_2, an isotropic gas phase silicon etchant is then used to etch the silicon substrate through the parylene openings. As a result, continuous trenches are formed in the silicon substrate via the discrete openings.

- Further comes the sealing of trenches to finish up the microchannel fabrication. This is achieved by depositing another layer of parylene C The parylene deposition is a very conformal process. Therefore, this parylene layer is able to deposit on the inside surface of the trench, forming bottom and side walls of the parylene channels. This step marks completion of step (4) in figure 3. An SEM image of cross section of the parylene microchannel is presented as figure 4.

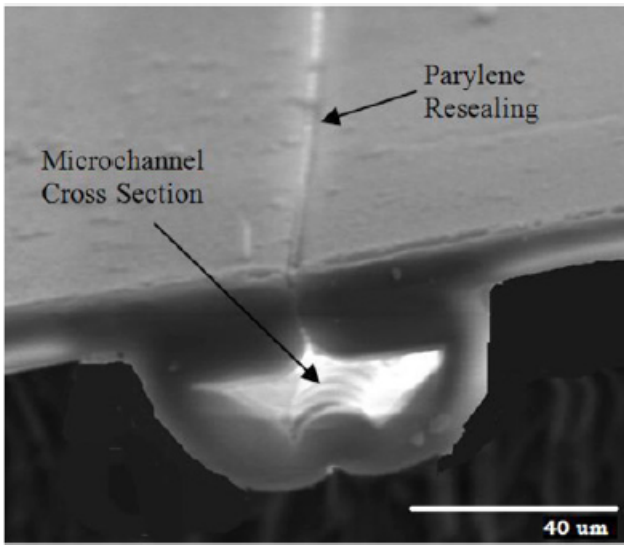


Fig 4:EM image of cross section of the parylene microchannel

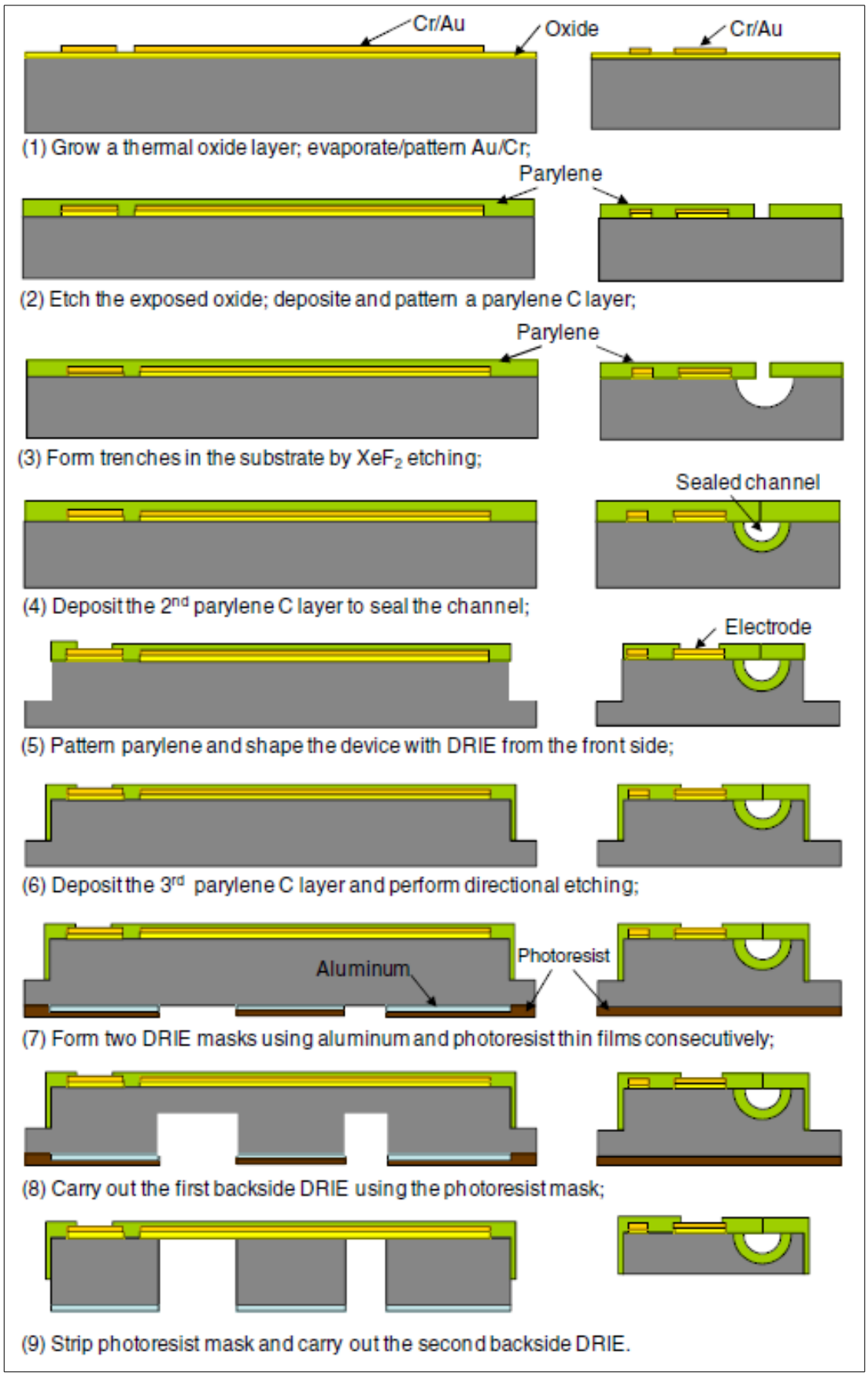


Figure 3: Simplified fabrication process of neural probes with integrated microchannels. Left column: cross-sectional view of silicon islands. Right column: cross-sectional view of one probe shank.

- The two parylene layers together are patterned using oxygen plasma with an aluminium mask
  in an effort to expose the desired
  electrode sites in addition to defining the front-side shapes of the devices.
  Next, DRIE is carried out from the front side of the Si wafer using the same aluminium mask until an etching depth of about 100 μm is reached (figure 2(5)). This etching is performed to shape the device, including the probe shanks and the device borders, from the front side

- A third parylene layer is deposited on the wafer through sputtering. This ensures that the silicon probe shanks can be encapsulated with parylene thin films on the sidewalls to prevent the peeling off of the top parylene layer during implantation. Moreover, the electrical insulation of metal traces is achived.

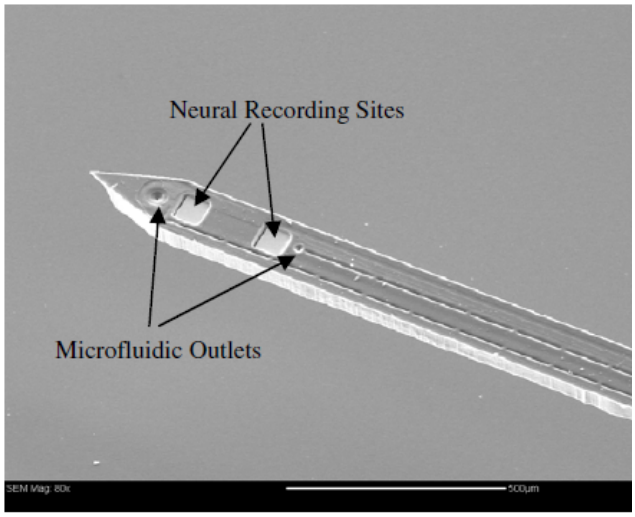


Fig 5: Shank with 2 electrodes and 2 microchannel outlet ports

- Next, we turn towards developing the back side of the wafer to realize silicon islands and probe shanks. DRIE is carried out on the backside of the wafer in this end. Two different masks are used to carry out this etching to generate different etching depths for islands and shanks. The first mask for silicon islands is made using aluminium. The second mask , which helps to release the flexible cables can be a thick photoresist.

- The first backside DRIE is carried out with an etch depth slightly larger than the front-side depth (110μm). The photoresist mask is then removed, and subsequently the second DRIE is performed using the aluminium mask to completely release the silicon islands and probe shanks.

Suggested changes:

The common output bus in the design under scrutiny was derived from one of the extreme islands. On the contrary, we propose deriving these channels from the middle island. This would improve (though minutely) the response time for the last island. Also, the delay between first island and the middle island will match the delay between last island and middle island which could ease synchronisation.

<u>Design 2</u>: Planar 2D-Array Neural Probe for Deep Brain Stimulation and Recording (DBSR)

Salient features of this design:

- The probe supports a 2D- array of small electrodes for selective and targeted interaction on the level of single neurons.

- The pixels in the design can be used for recording impulses from neurons as well as exciting them depending on the applications.
- In this design, the amount of available electrode sites was however limited to five.
- We can increase the probes thus it is fully scalable.

Abstract of design for the 3D probe presented henceforth:

The fabricated device shown consists of a 2-D electrode array where each pixel functions either as a recording site or a stimulation site. The geometric area is as small as possible to allow accurate recording or stimulation. This will enable recordings of the neural activity with high spatial resolution and high signal to noise ratio.

An assortment of passive planar silicon probes with 2D- arrays of varying electrode configurations has been fabricated. Four different probe configurations include equidistantly arranged disk-shaped electrodes with diameters ranging from 50 to 4 µm.

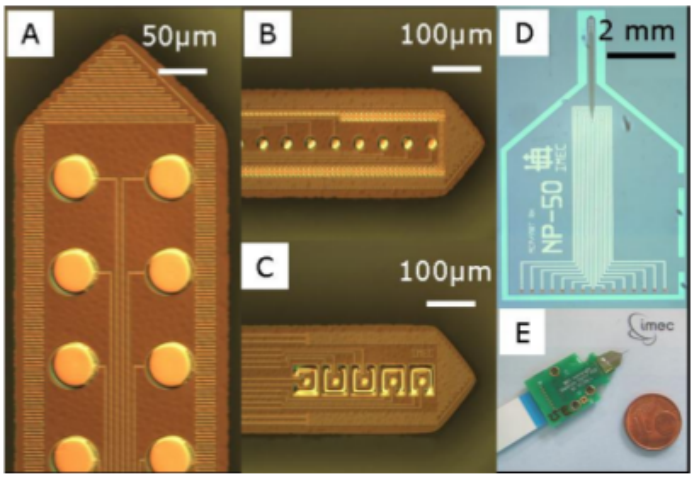


Fig. 1. Optical micrographs of probe tips with different electrode sizes and configurations: tip with electrodes of 50 µm diameter (A), of 25 µm diameter (B) and with C-shaped electrodes (C). Picture of the whole probe (D) and the packaged probe (E).

Another probe design contains C-shaped electrodes each with an enclosed disk- shaped reference electrode. Probes with a 2mm long shaft and 200×200 µm 2 cross- section have been realized.

Fabrication Process:

- The first step defines the metal interconnects and the temperature sensor by means of lift-off of a metal stack of titanium-tungsten/gold/titanium (30 nm/100 nm/30 nm). Titanium-tungsten (TiW) was used for metal adhesion, while the additional titanium (Ti) layer prevented direct exposure of the gold in subsequent gold-free processing tools.
- Next, thin tantalum pentoxide (Ta_2O_5 , 100 nm) was deposited as adhesion layer for 1 µm of Parylene C which was deposited and polymerized by vapor condensation.

- Reactive ion etching (RIE) through a photoresist etching mask was then applied to the Parylene C and Ta_2O_5 to open windows for the electrodes and bond pads.

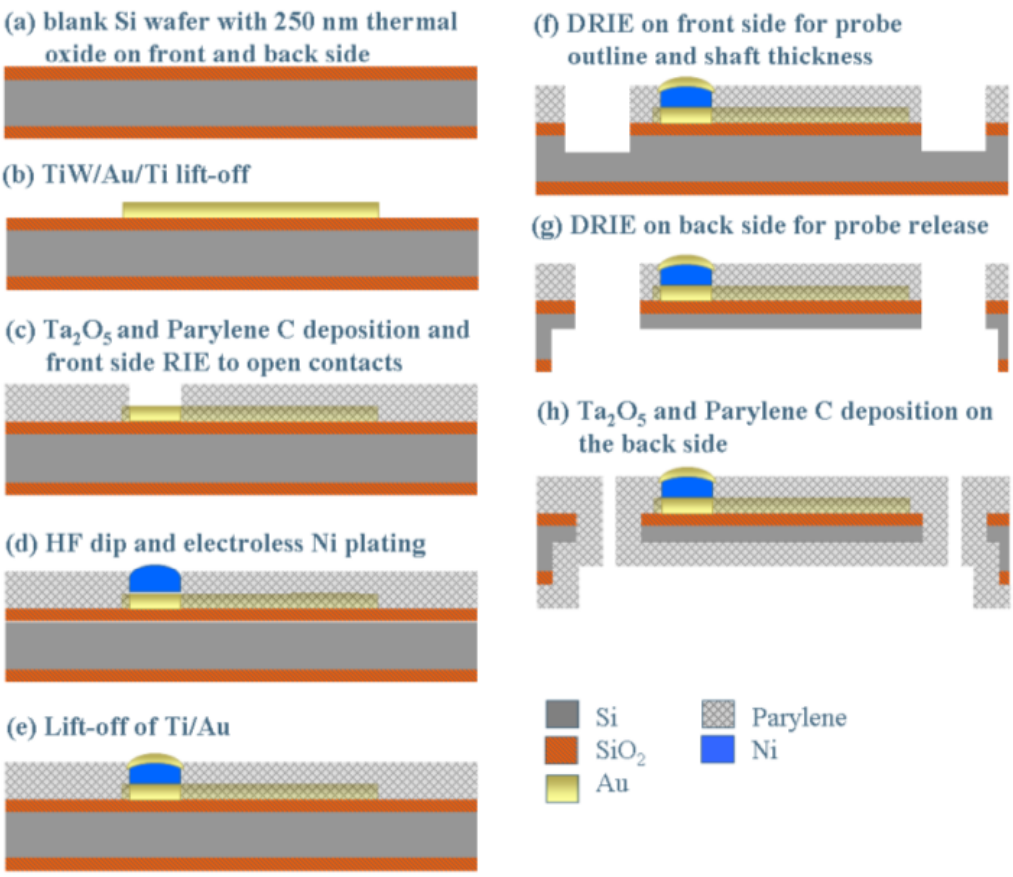


Fig. 2. Schematic of the process flow.

- After removing the Ti from the Au contacts with a short dip in diluted hydrofluoric acid (HF), the Parylene C openings were filled with around 1 µm nickel (Ni) by electroless plating. This additional plating step resulted in electrodes with a curved rather than planar surface, which is expected to generate a more homogeneous current density distribution during stimulation.

- In another lift-off step the Ni electrodes were covered with a stack of Ti/Au (30 nm/300 nm) for improved biocompatibility.

- Using a thick patterned resist on the front side as an etch mask, the probe outline and shank thickness were then defined by first dry etching the Parylene C, Ta_2O_5 and SiO 2 and subsequent deep reactive ion etching (DRIE) of 200 µm deep silicon trenches. A second DRIE step was applied to the back side to etch through the wafer while leaving two struts as breaking points for easy probe release. 100 nm of Ta_2O_5 were then deposited on the backside of the wafer that was fixed on a carrier wafer for better handling. After taking the wafer from the carrier, 1 µm of Parylene C was evaporated on the back side while keeping the front side protected with blue dicing tape.

<u>Design 3</u>: 3D flexible multichannel neural probe array

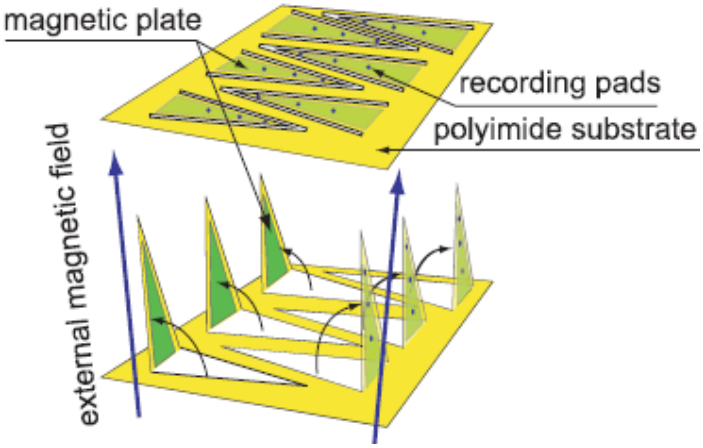
In the past decade we have mostly used silicon structures for the insertion probes because

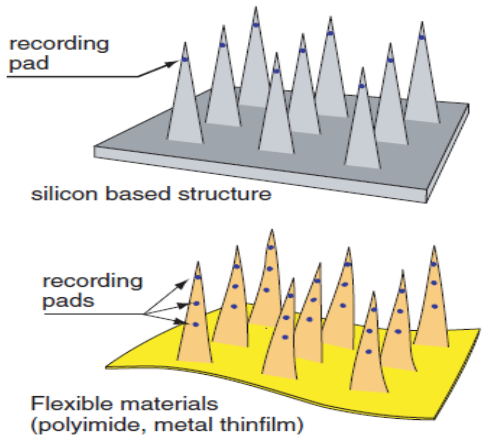
they can be fabricated to a small scale. One major drawback of using silicon is that it **lacks deformity.** This is why such kind of rigid electrode is not suitable for long-term recording. There has been an increasing demand for **flexible electrodes**. Such devices can be employed by using flexible materials like polyimide, SU-8, parylene and PDMS. Using these kind of flexible electrodes, one can attain implantation with minimum damage to the neural tissues.

Salient Features:

- Probes are made up of **flexible** material (polyimide, being used here)
- Probes are bent using **external magnetic field** to achieve the desired position.
- Can be used to measure brain activity at **various depths** simultaneously due to several recording pads on each probe.
- Able to achieve **multichannel** neural recording from three-dimensional (3D) positions.

Abstract for the design of 3D flexible multichannel neural probe array:





As shown in the figure, traditional electrode have only one single recording channel at a probe made of silicon. Here we propose a device which has **three recording pads** on a single probe which can measure the neural activity at various depths of the brain.

In the first figure below we have shown the magnetic batch assembly of the probes. After applying a magnetic field as shown, we observe the bending of the probes. To bend the probes at an angle of 90 degrees we have to apply a certain value of magnetic field.

Here we propose a device with **2 x 3** probe arrays and **18** neural recording channels.

Fabrication Process:

- **Electroplating nickel on a silicon wafer (Refer Fig. a):** We use a 250 um thick silicon substrate as the starting material. Nickel electroplating is done first to form the magnetic backing plate of each probe. The thickness of the nickel plate used is 5 um.

- **Spin-coating polyimide along with deposition and patterning of metal layers (Refer Fig. b):** A titanium layer is deposited on an aluminium layer with a vacuum evaporator to be used as the electrode pads and interconnections and patterned by wet etching. The total thickness of these metals is about 1 um.

- **Spin-coating polyimide again to cover the metal layer (Refer Fig. c):** On top of the

metals we spin coat another layer of polyimide. The two metal layers are sandwiched between two 10 μm polyimide layers. which are spin coated and patterned by O2 plasma for about 30 min.

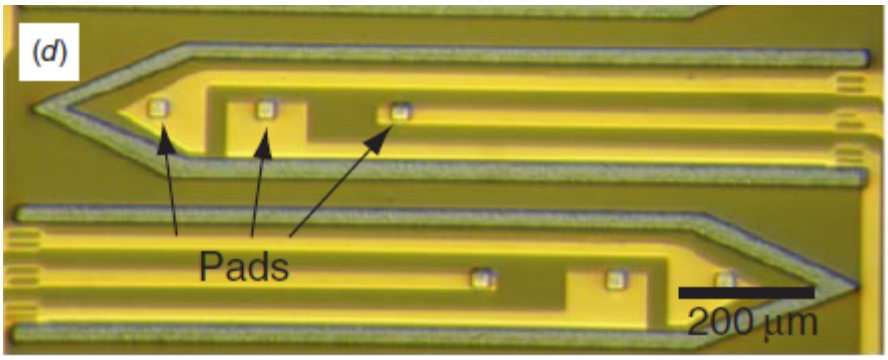
- **Patterning the polyimide by O2 plasma (Refer Fig. d):** The polyimide layers are spin coated and patterned by O2 plasma for about 30 min. During this patterning, an aluminium layer is deposited and patterned on the polyimide layer to be used as a mask to define the electrode outline, bonding regions and recording pads. After this process, we form the through holes, and the titanium layer appears. These through holes are formed on each end of the interconnecting cable, so that only this area of the cable would contact the neural tissues. The aluminium mask layer is removed with an etchant that did not attack the titanium layer.

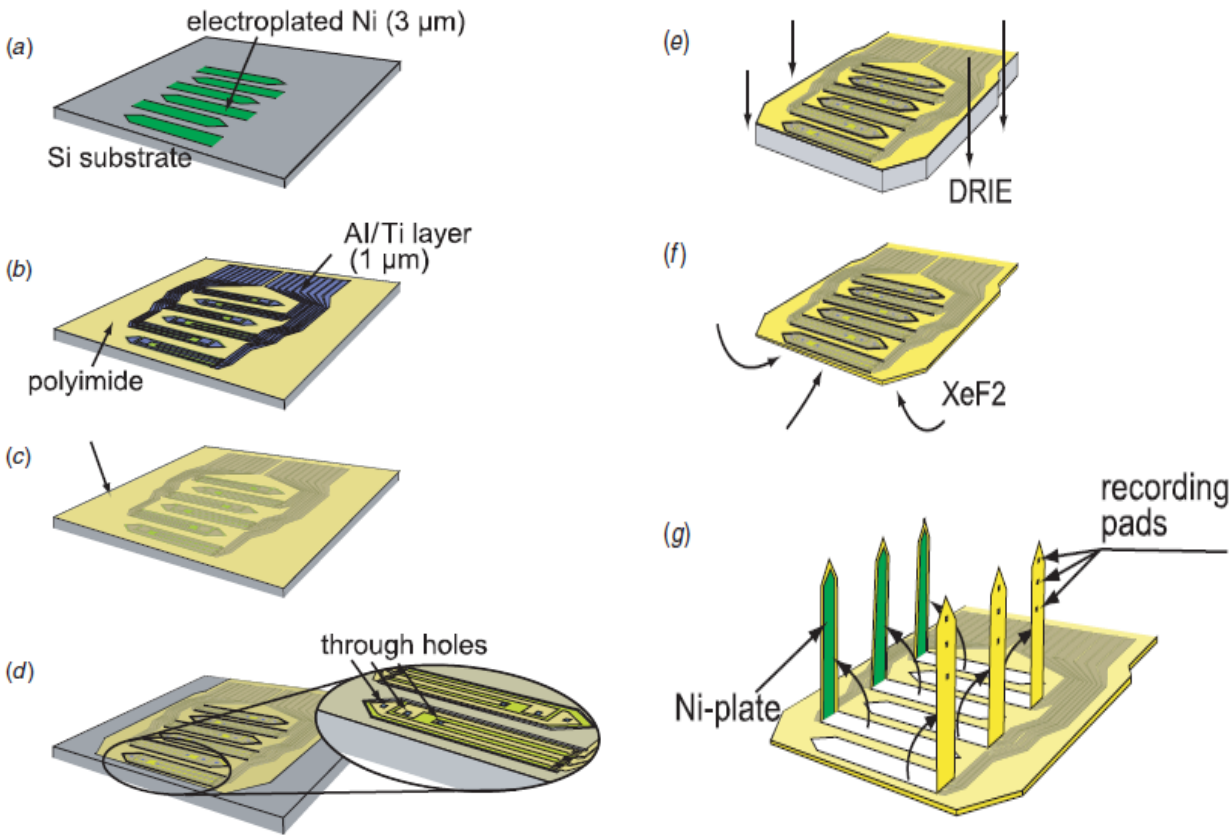
- **Etch the silicon from top and backside by Deep Reactive Ion Etching (DRIE) (Refer Fig. e):** We use deep reactive ion etching (DRIE) and XeF2 etching processes to remove the silicon substrate. We first use DRIE to etch about 20 μm from the top, and then again from the backside.

- **Removing the residual silicon on the backside of the electrode by XeF2 (Refer Fig. f):** Finally, the residual silicon is isotropically etched by XeF2 to produce the flexible electrode whose total thickness is about 25 μm.

- **Folding the probes (Refer Fig. g):** The probes are vertically aligned by applying an external magnetic field.

- **Dimensions of the probe:** Each probe is 1.2 mm in height and 160 μm in width and has three 20 × 20 μm2 recording pads at 200 μm intervals (figure below). The precise depths of each recording pad when implanted were 0.65 mm, 0.85 mm and 1.05 mm





Customer Requirements:

4. High sensitivity towards neural impulses
5. Low cost of fabrication
6. High resolution
7. Immunity to shocks and vibration
8. Convenient interfacing with neurons, with minimum inflammation (biocompatibility)
9. Minimum neural damage during piercing
10. Neuron density around the electrode must remain unaffected
11. Long operational life
12. Easy interfacing with external circuitry
13. Small size and light weight

Functional Specifications:

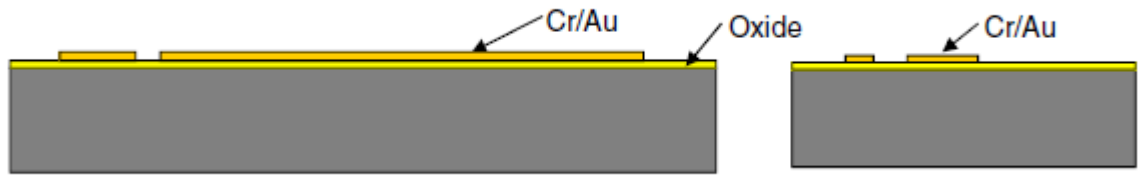
Metric	**Details**	**Customer Requirements Fulfilled**
1. Low Cost	- Replication of islands, reducing bulk production cost, permitted by symmetric structure	2
2. Suitability for biomedical applications	- The whole device is coated with a parylene C layer, which is widely employed due to its biocompatibility.	5, 7

3. Flexible interconnect cables	- Probe islands are connected to bonding pads via a flexible parylene interconnect which improves its performance againt stresses - Islands are interconnected through folds using similar flexible interconnected	4, 9
4. Biocompatable electrode coating	- Coating electrode sites with a nona-peptide along with polypyrrole and laminin facilitates neural signal transport along with improved improved neural density around the electrodes.	1, 7
5. Triangular shanks	- Thicker portion of the shanks connected to the islands to be made triangular to reduce damage during piercing	6
6. Fully coated device	Parylene coating all over the device - Removes the problem of shorting conducting buses - Improves robustness of probe tips by absorbing stress due to deformation caused by piercing - Eliminates chances of corrosion of metal/silicon surface	1, 4, 8

Fabrication process:

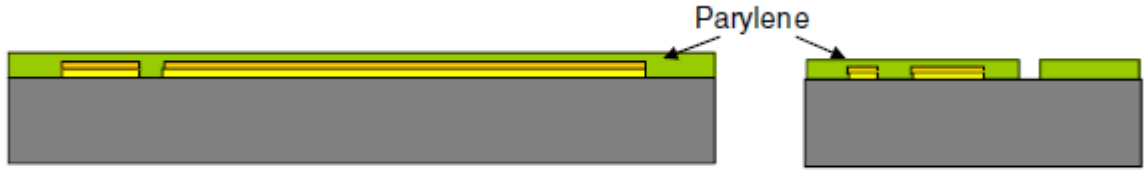
1. Fabrication process starts with the thermal oxidation of 300nm silicon oxide layer on a double side polished 512um thick 5 inch silicon wafer. This oxide layer serves as an insulator for the metal electrode.

2. Then a chrome (20nm) and gold(200nm) layer are deposited sequentially on top of SiO2 and then they are patterened. Surface micromachining is used for the deposition in which a photoresist is spin coated on the wafer and then a photomask with desired patterns is aligned on it ,further the system is exposed and developed to generate the desired pattern. Now the chrome and gold layers are etched using gold and chrome etch bath respectively, then residual photoresist is dissolved to get the final patterned Cr and Au.



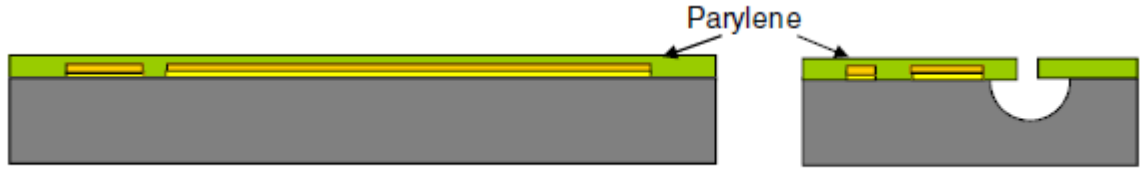
(1) Grow a thermal oxide layer; evaporate/pattern Au/Cr;

3. Now, the exposed silicon dioxide layer is etched by Drytek reactive ion etching (DRIE) using C6F6. A 5 µm thick parylene C layer is deposited and patterned to open to open the access areas for the microchannels (including inlet port and outlet ports). The layer is patterned using oxygen plasma etching with the help of an aluminium mask.



(2) Etch the exposed oxide; deposite and pattern a parylene C layer;

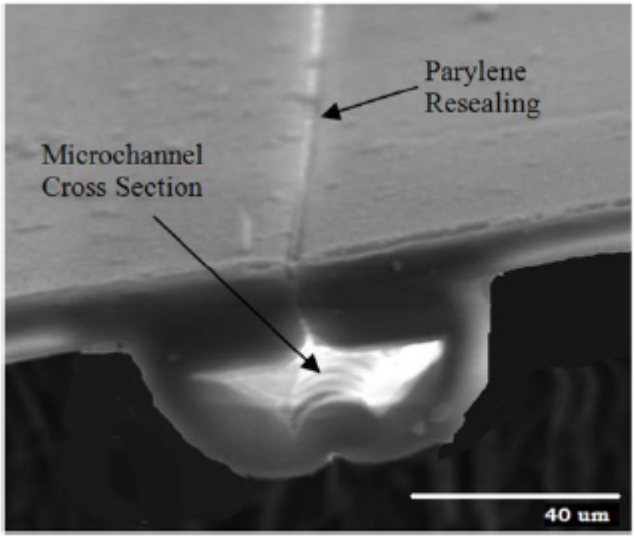
4. XeF2, an isotropic gas phase silicon etchant is then used to etch the silicon substrate through the parylene openings. As a result, continuous trenches are formed in the silicon substrate via the discrete openings.



(3) Form trenches in the substrate by XeF_2 etching;

5. Then comes the sealing of trenches to finish up the microchannel fabrication. This is achieved by depositing another layer of parylene C The parylene deposition is a very conformal process. Therefore, this parylene layer is able to deposit on the inside surface of the trench, forming bottom and side walls of the parylene channels.

(4) Deposit the 2nd parylene C layer to seal the channel;



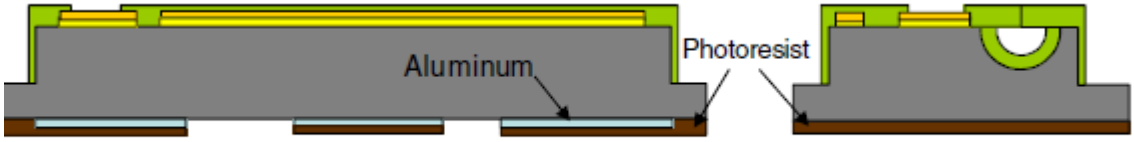
An SEM(scanning electron microscope) image of cross section of the parylene microchannel

6. This parylene layer, together with the first one, is patterned using oxygen plasma with an aluminum mask in an effort to expose the desired electrode sites in addition to defining the front-side shapes of the devices. next, DRIE is carried out from the front side of the Si wafer using the same aluminum mask until an etching depth of about 100 μm is reached.This etching is performed to shape the device,inlcuding the probe shanks and the device borders, from the front side.

7. A third parylene layer is deposited on the wafer through sputtering. This ensures that the silicon probe shanks can be encapsulated with parylene thin films on the sidewalls to prevent the peeling off of the top parylene layer during implantation. Moreover, the electrical insulation of metal traces is achieved.

(6) Deposit the 3rd parylene C layer and perform directional etching;

8. Next, we turn towards developing the back side of the wafer to realize silicon islands and probe shanks. DRIE is carried out on the Fig 5: Shank with 2 electrodes and 2 microchannel outlet ports backside of the wafer in this end.
Two different masks are used to carry out this etching to generate different etching depths for islands and shanks. The first mask for silicon islands is made using aluminium. The second mask ,which helps to release the flexible cables can be a thick photoresist.



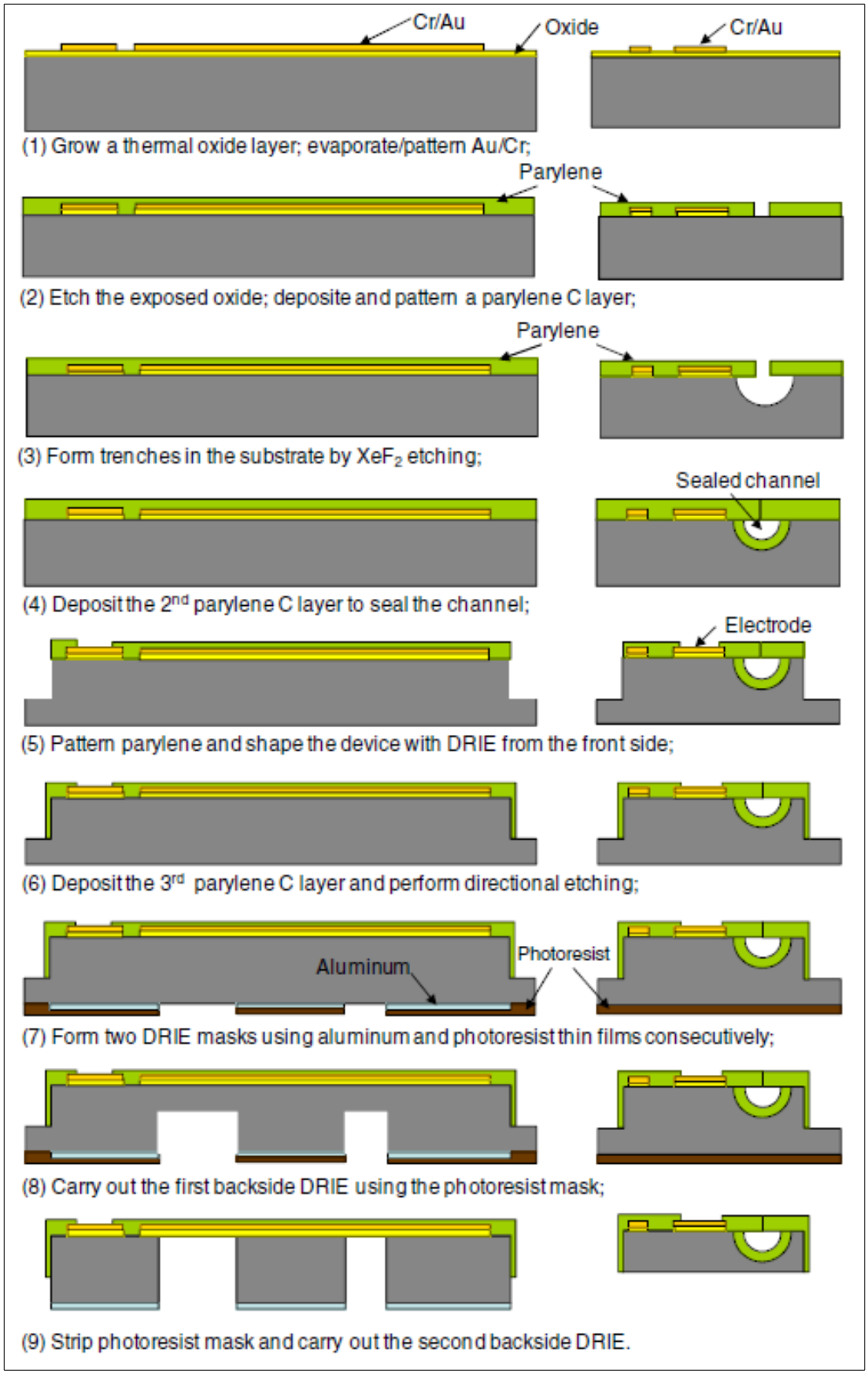
7) Form two DRIE masks using aluminum and photoresist thin films consecutively;

9. The first backside DRIE is carried out with an etch depth slightly larger than the frontside depth (110µm). The photoresist mask is then removed ,and subsequently the second DRIE is performed using the aluminium mask to completely release the silicon islands and probe shanks.

(9) Strip photoresist mask and carry out the second backside DRIE.

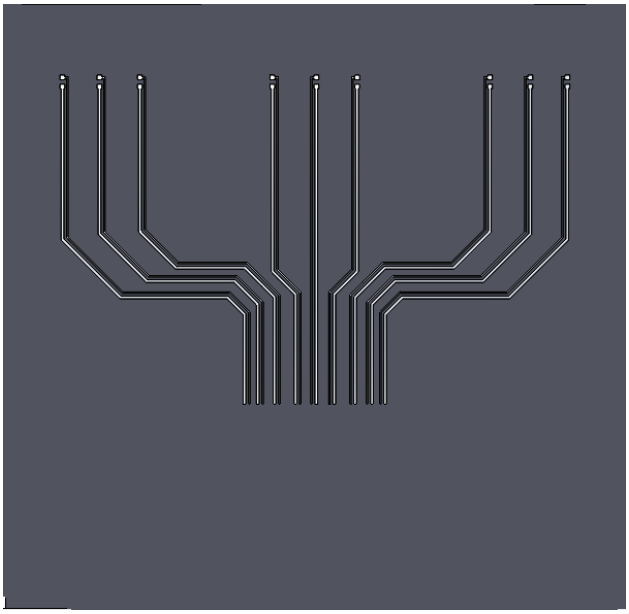
Mask Designs:

Steps involved in the farication process can be summaized in the following diagram:

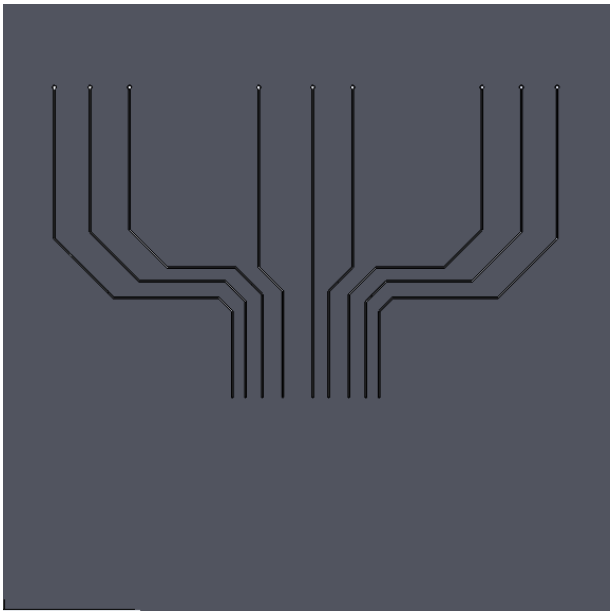


Masks need to be used in steps 1, 2, 4 and 7.

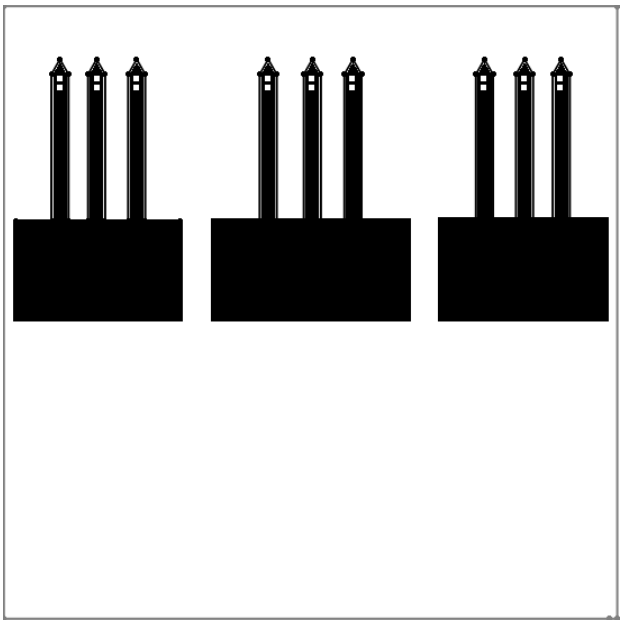
Step 1: Resist Type – Negative



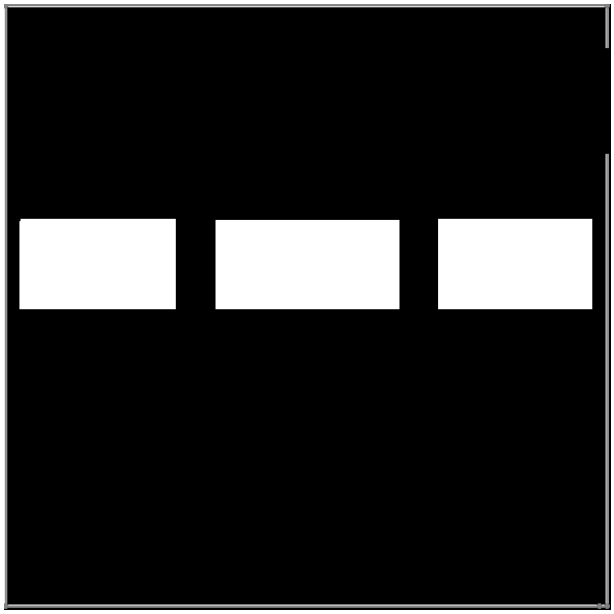
Step 2: Resist Type – Positive



Step 4: Resist Type – Positive



Step 7: Resist Type – Negative



Packaging:

The packaging serves to integrate all of the components required for a system application in a manner that minimizes size, cost, mass and complexity. The packaging provides the interface between the components and the overall system. The following subsections present the two main features involved in MEMS packaging: connection among the different system components,mechanical support and protection from the environment.

1. Connection among the different system components:

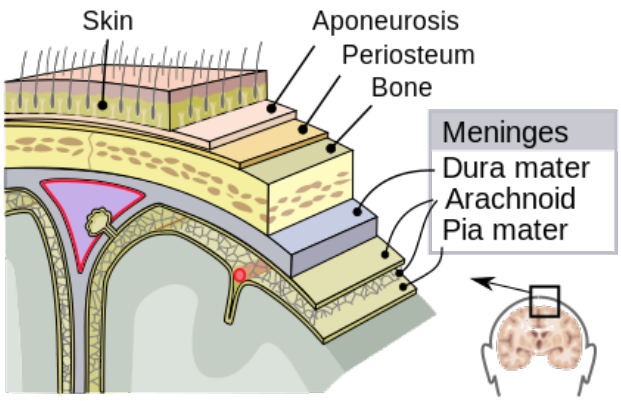
Our design contains several components, one in which all the three islands are attached to each other with flexible parylene cable then there are structures of shanks for each island and finally there is the interface die with bonding pads which is connected by a flexible cable. Now all these components are glued together by using epoxy to form the complete assembly and to ensure electrical connections between constituents wires run from one structure to another.

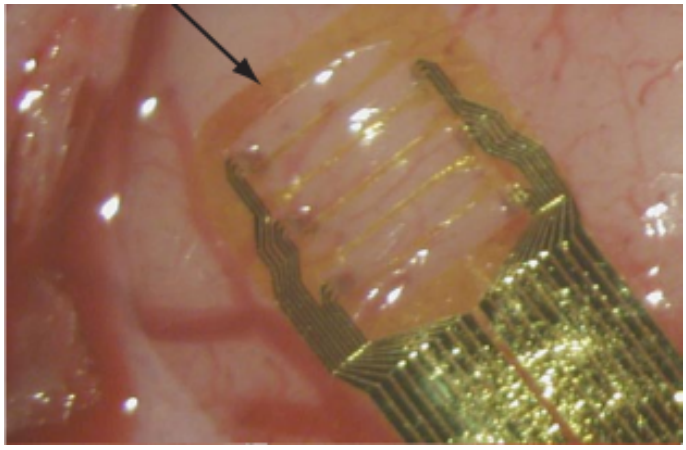
2. Mechanical support and protection from the environment:

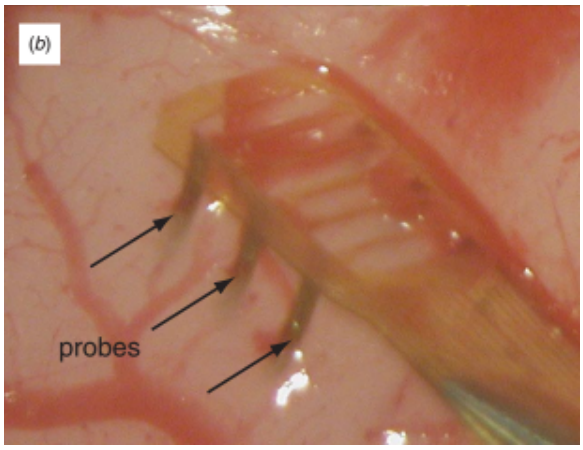
Due to the nature of MEMS being mechanical, the requirement to support and protect the device from thermal and mechanical shock, vibration, particles, and other physical damage during storage and operation of the part must be addressed also many elements in the environment can cause corrosion or physical damage to the metal lines of the MEMS as well as other components in the package thus to solve these problems the whole system is encapsulated inside a material and as our device will be used for in-vivo implantation thus the material for encapsulation must be biocompatible and the design of this material will be such that there will be space provided for the shanks to protrude out of it so that the sensing and stimulus of electrical signals can take palce.

Testing:

We can use an anesthetized rat (Laboratory rat) for the neural recording experiment for this probe array. It has to be inserted into the rat's cortex after peeling off the dura mater, which is a tough fibrous membrane covering the cortex. The probes are easily inserted into the organs at an angle, when the island structures (including the shanks) are pushed with tweezers under a microscope. We should not observe probes bending back during the insertions. Here we have used parylene as it has good biocompatibility and is easy to combine with our fabrication process. After doing this, the flexible substrate fitted into the brain surface well. Using this electrode, we successfully measure the neural activity from the rat's brain. The neural signals are amplified by an amplifier. These signals can then be recorded on a PC by way of an Analog-Digital converter.







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